Maxim Abdulkhalikov

Software Developer | React, TypeScript, Next.js | Winnipeg, MB
204-996-2313 | abdulhalikov0@gmail.com | linkedin.com/in/abdulkhalikov | github.com/maxakadem1

Summary

Software developer and acting technical lead building the platform behind lottery products used by millions of players across 8 clients. Owning systems from architecture to production, leading multi-developer launches, and mentoring engineers.

Experience

Software Developer, Core Product Team Mar 2024 – Present
Pollard Banknote *Winnipeg, MB*

- Own the shared platform behind Pollard's lottery products: client web and mobile apps plus an embeddable widget system running 40+ widgets in production for each of 8 clients across multiple environments, serving millions of players.
- Architected the widget system and 30+ React components, building about half of them myself.
- Acting technical lead for Arizona's relaunch: led 8 developers over 12 weeks, organized workstreams, assigned ownership, and unblocked the team; shipped on time, with 30,000+ players on the site immediately after launch.
- Developed react-core, the shared TypeScript API layer connecting every front end to Pollard's backend services for authentication, loyalty points, location verification, and gameplay.
- Built Konsole, an internal tool for creating and managing database records (games, polls, surveys) through APIs instead of direct database edits, with role-based permissions and validation workflows.
- Built a standalone widget development environment (sandboxes, hot reloading, Storybook, mock APIs, test harnesses) so developers and QA test widgets in isolation, cutting QA time per widget from a full day (7–8 hours) to under 10 minutes.
- Chose and led the widget build's migration from Create React App to Vite, cutting dependency vulnerabilities from 380 (25 high/critical) to 0, shrinking bundles, and making the codebase maintainable again.
- Led a production app's migration to Next.js, improving Speed Index from 5.1s to 1.3s and Largest Contentful Paint from 4.2s to 2.0s.
- Built the build, versioning, and publishing pipelines for shared packages; deploy production Next.js apps on AWS Amplify with Route 53 DNS.
- Technical escalation point for about 15 developers; mentor co-op through senior developers via pairing, code review, onboarding, and architecture documentation.

Frontend Architect and Developer May 2023 – Jan 2024
OutreachGenius *Winnipeg, MB*

- One of the first engineers at an early-stage startup; architected the product's UI system for both the SaaS web app and the Chrome extension.
- Rewrote the Chrome extension from a legacy Vue codebase into a maintainable React and TypeScript codebase; the extension grew from 0 to 500+ users.
- Established a Tailwind-based design system shared by the web app and extension, giving both products one consistent set of components and styles.
- Integrated Firebase for real-time data and authentication across the web app and extension.
- Designed the SaaS and extension layouts in Figma and implemented them end to end.

Projects

Godot Docs RAG | *Python, FastAPI, OpenAI, Pinecone, Next.js* | github.com/maxakadem1/godot-rag 2026

- Built a RAG system over the Godot Engine docs (1,565 pages, 8,580 chunks) on the OpenAI and Pinecone SDKs, with hybrid vector and keyword search and answers that cite their sources.
- Built an evaluation set of 38 labelled questions to measure retrieval: vector search reached 89% hit rate@5 and doubled recall on paraphrased questions over keyword search (30% to 60%).
- Deployed a FastAPI backend and Next.js frontend on Vercel with streamed responses, rate limits, and a daily cap on LLM cost.

Visual Lab | *Next.js, React, TypeScript, WebGL, Canvas 2D* | visual-lab-chi.vercel.app | GitHub

- Designed and built a browser-based image and video editor with 10 customizable effects (pixelate, dither, bloom, glitch, and more) that run in real time, entirely client-side.

- Built a layer system for stacking, drag-to-reorder, configuring, hiding, and removing effects, with live previews as parameters change.
- Implemented two rendering pipelines: Canvas 2D for images and WebGL shaders for video, so effects stay real-time on moving footage.
- Added palette extraction, drag-and-drop uploads, video playback controls, and full-resolution PNG export.

Technical Skills

Languages: TypeScript, JavaScript, Java, GraphQL, Python
Frameworks: React, Next.js, React Native, Node.js, Vite, Vue.js
Cloud and Infrastructure: AWS (Amplify, Route 53), Firebase, Vercel, CI/CD, monorepos, package publishing
Tools: Jest, Playwright, Storybook, Contentful (CMS), Redux, Tailwind, Figma, accessibility (WCAG)

Education

University of Manitoba — Winnipeg, MB
B.Sc. Computer Science — *Sep 2018 – May 2023*

Art School of Kasteyev — Almaty, KZ
Diploma in Art and Design — *Sep 2016 – Jun 2018*

Volunteering

DevClub / Hackathon Winnipeg | *Promotions Executive, Lead Designer* — Apr 2022 – Present
- Organized the largest UofM hackathon since 2015; secured sponsors for events of 200–300+ students.